

Complications associated with pulsed field ablation vs radiofrequency catheter ablation of atrial fibrillation

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ABSTRACT

BACKGROUND The first commercial systems for pulsed field ablation (PFA) were recently approved by the Food and Drug Administration, but their safety profile in nationwide practice remains underexplored.

OBJECTIVES This study aimed to describe reported adverse events with PFA systems after their market release and to compare the relative proportion of complications associated with PFA vs radiofrequency (RF) ablation catheters.

METHODS We conducted a search of the Manufacturer and User Facility Device Experience database for adverse event reports involving Food and Drug Administration–approved PFA and RF catheters from January 1, 2024, to July 31, 2024.

RESULTS A total of 1237 reports detailing clinical complications ($n=156$ for PFA and $n=315$ for RF) or catheter malfunctions ($n=336$ for PFA and $n=430$ for RF) were analyzed. The most common adverse events reported for PFA were pericardial effusion, vasovagal response, and hemolysis, compared with pericardial effusion, ischemic stroke, and esophageal damage for RF ablation. The proportion of reported adverse events that were deaths was lower with PFA than with RF (2.6% vs 8.9%; $P=.010$). Hemolysis (9.0% vs 0%), coronary events (5.8% vs 0.6%), and vasovagal responses (14.1% vs 0%) were reported more frequently after PFA ($P<.001$ for all). In contrast, esophageal damage (0% vs 4.1%; $P<.001$) and pulmonary vein stenosis (0% vs 1.9%; $P=.184$) were not reported with PFA, but only with RF. Regarding catheter malfunctions, mechanical problems were reported in higher proportion after PFA than after RF (87.2% vs 17.2%; $P<.001$).

CONCLUSION The reported complications associated with PFA and RF ablation differ significantly. Close attention to PFA-specific complications is essential for ensuring the safe use of PFA and for mitigating risk.

KEYWORDS Atrial fibrillation; Outcomes; Ablation; Complication; Electroporation; Radiofrequency ablation

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Introduction

Pulsed field ablation (PFA) is now approved in the United States, and its use is increasing in clinical practice. This novel energy source delivers repetitive, short, high-voltage electric pulses that create micropores in plasma membranes, irreversible electroporation, and cell death.¹ PFA demonstrates tissue selectivity, with myocardial cells being more sensitive than smooth muscle or nerve tissues. Theoretically, PFA could be an ideal energy source, capable of creating transmural,

irreversible lesions rapidly without permanently damaging adjacent tissues.^{2–5} The efficacy of PFA has been evaluated in prospective clinical trials, demonstrating comparable efficacy to conventional ablation modalities with shorter procedure times and similar rates of adverse events.^{6–10} However, after its introduction into clinical practice, infrequent yet unexpected complications have been reported,¹¹ including those observed with thermal ablation and some specific to PFA. Most of these reports come from

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individual practitioners or centers, and thus the generalizability of these findings is unknown. With the recent Food and Drug Administration (FDA) approval of PFA and its anticipated widespread use in the United States, it is crucial to understand the incidence and nature of complications encountered in general clinical practice. Therefore, we sought to describe complications associated with PFA in US clinical practice as reported in the Manufacturer and User Facility Device Experience (MAUDE) database. We also hypothesized that ablation using PFA catheters would result in a different complication profile compared with traditional radiofrequency (RF) ablation.

Methods

The FDA MAUDE database is an open database containing medical device reports submitted by device manufacturers and physicians. We initially queried MAUDE for all adverse event reports related to medical devices using PFA and open-irrigated RF ablation catheters currently used in clinical practice by using their brand names (Table 1). We excluded reports that (1) were duplicates, (2) came from a literature source that did not include an individual patient report, (3) involved catheters that were not used for atrial fibrillation (AF) or pulmonary vein isolation, as detailed in the MAUDE database, or (4) reported recurrence of atrial arrhythmia (fibrillation or flutter) as a complication.

All medical device reports were initially classified as a “clinical complication” (including event type of “death” or “injury”) or “catheter malfunction” (including event type of “malfunction”). These reports were subsequently manually reviewed and subcategorized by 2 experienced electrophysiologists (M.S.C. and J.P.P.) according to the pathophysiology of injury/death as well as mechanism catheter malfunction. Cases with discordance were resolved by consensus. Since the MAUDE database is a de-identified and publicly accessible database, informed consent or institutional review board approval was not required for this study.

The proportion of adverse events in each category was presented as counts and percentages. Between-group comparisons were made using the χ^2 test or Fisher exact test, as appropriate, and 2-sided *P* values < .05 were considered significant. All statistical analyses were performed using R version 4.3 (R Foundation for Statistical Computing, Vienna, Austria).

Abbreviations

AF: atrial fibrillation

AVB: atrioventricular block

FDA: Food and Drug Administration

MAUDE: Manufacturer and User Facility Device Experience

PFA: pulsed field ablation

PPM: permanent pacemaker

RF: radiofrequency

SND: sinus node dysfunction

Results

During the study period from January 1, 2024, to July 31, 2024, a total of 2093 MAUDE reports of complications after the use of PFA (n=542) or RF (n=1551) were identified (Figure 1). After exclusions, 1237 reports on clinical complications (n=471) or catheter malfunctions (n=766) were

analyzable. Among the analyzable reports, 1227 (99.2%) were manufacturer reports, 8 (0.6%) were voluntary reports, and 2 (0.2%) were user facility reports.

Adverse events reported with PFA

For those categorized as clinical complications, 156 reports (152 injury and 4 death) after PFA and 315 reports (287 injury and 28 death) after RF were available (Figure 2 and Online Supplemental Table S1). The most common reported clinical complication after PFA use was pericardial effusion (n=53), 88.7% (n=47) of which required pericardiocentesis or surgical drainage. The second most common report after PFA use was vasovagal response (n=22), with all cases resolving through either medication (atropine or isoproterenol; n=11) or temporary pacing (n=11). Hemolysis was reported in 14 cases, with acute renal failure occurring in 9 cases, and 4 of those patients ultimately required dialysis. All cases of hemolysis were reported after the use of the Farawave catheter, with none occurring after the use of PulseSelect catheter. Coronary events were reported in 9 cases, all attributed to coronary vasospasm that did not require percutaneous intervention. Most coronary events (n=7) occurred intraprocedurally, with 2 occurring postprocedurally. One patient experienced incessant ventricular tachycardia and fibrillation associated with ST-segment elevation, which resolved with nitroglycerin administration.

There were 9 cases of bradyarrhythmia, including 3 cases of atrioventricular block (AVB) and 6 cases of sinus node dysfunction (SND), all of them resolved without the need for permanent pacemaker (PPM) implantation. Ischemic stroke and phrenic nerve damage were reported in 9 and 2 cases, respectively. Other complications reported after PFA included heart failure (n=6), hypotension (n=5), respiratory disorders (n=5), bleeding (n=4), death from unknown causes (n=3), air embolism (n=3), renal impairment (n=2), tissue damage (n=2), pain (n=2), sedation issues (n=2), femoral vein injury (n=1), vomiting (n=1), fever (n=1), and left atrial thrombus (n=1).

Adverse events reported with RF ablation

The most common adverse events reported with RF were pericardial effusion (n=225), of which 95.6% (n=215) required pericardiocentesis. Ischemic stroke was reported in 20 cases, making it the second most reported complication. There were 13 cases of esophageal damage, 46.2% (n=6) of which resulted in death. Bradyarrhythmia was reported in 9 cases (8 AVB and 1 SND), with 5 cases of AVB requiring PPM implantation. Two coronary events were reported, including one case requiring thrombectomy and stenting and the other attributed to coronary spasm. Phrenic nerve damage and pulmonary vein stenosis were reported in 7 and 6 cases, respectively. Other complications reported with RF use included heart failure (n=6), death from unknown causes (n=6), hypotension (n=5), respiratory disorders (n=5), bleeding (n=3), ventricular fibrillation (n=2), neurostimulator dysfunction (n=1), coronary sinus dissection (n=1), air embolism (n=1), lead dislodgment (n=1), esophageal damage during

Table 1 Profile of the included ablation catheters

Brand name	Manufacturer	Energy source	Catheter diameter (F)	MAUDE reports
Farawave, Farapulse	Boston Scientific	Pulsed field	12	438
PulseSelect	Medtronic	Pulsed field	9	54
ThermoCool SmartTouch	Biosense Webster	Radiofrequency	8	45
ThermoCool SmartTouch SF	Biosense Webster	Radiofrequency	8	374
QDOT Micro	Biosense Webster	Radiofrequency	8	256
TactiCath	Abott	Radiofrequency	8	41
Flexability	Abott	Radiofrequency	8	2
IntellaNav MiFi	Boston Scientific	Radiofrequency	7.5	14
IntellaNav StablePoint	Boston Scientific	Radiofrequency	7.5	13

transesophageal echocardiography (n=1), and unintended left atrial appendage isolation (n=1).

Reports of catheter dysfunction

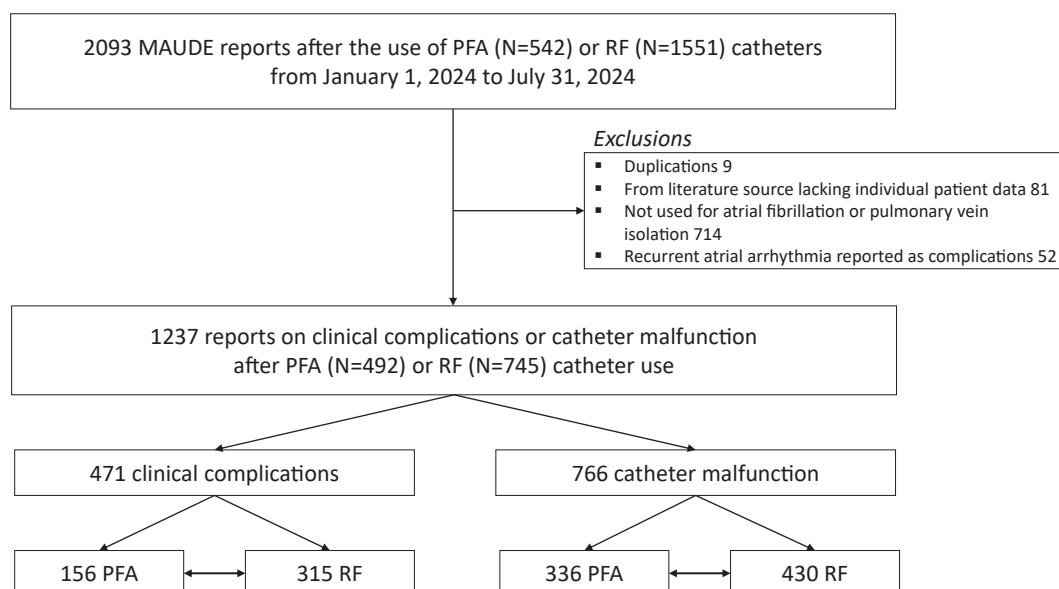
For those categorized as catheter malfunctions, 336 reports after PFA and 430 reports after RF were available (Figure 3 and Online Supplemental Table S2). The most common reported catheter malfunction for PFA use was mechanical dysfunction (n=293). Among mechanical issues reported with the Farawave catheter (n=258), the guidewire becoming stuck within the catheter lumen (n=77) and failure to modify or maintain the desired spline configurations (n=67) were the most common causes. For the PulseSelect catheter (n=35), the inability to maintain array shape was the most frequent mechanical issue (n=24), with 10 cases reporting the array becoming knotted but able to be withdrawn through the sheath without vascular injury. There were 35 reports of irrigation problems, 4 reports of functional problems (4 electric current errors in the PulseSelect catheter), 3 reports of signal noise, and 1 report of a coagulum trapped within the spline.

The most common catheter malfunctions after RF use were reported as functional problems (n=191), including force- or temperature-sensing errors, display issues, and communication errors. Other catheter malfunctions included irrigation problems (n=89), mechanical problems (n=74), coagulum or char formation (n=53), and signal issues (n=23).

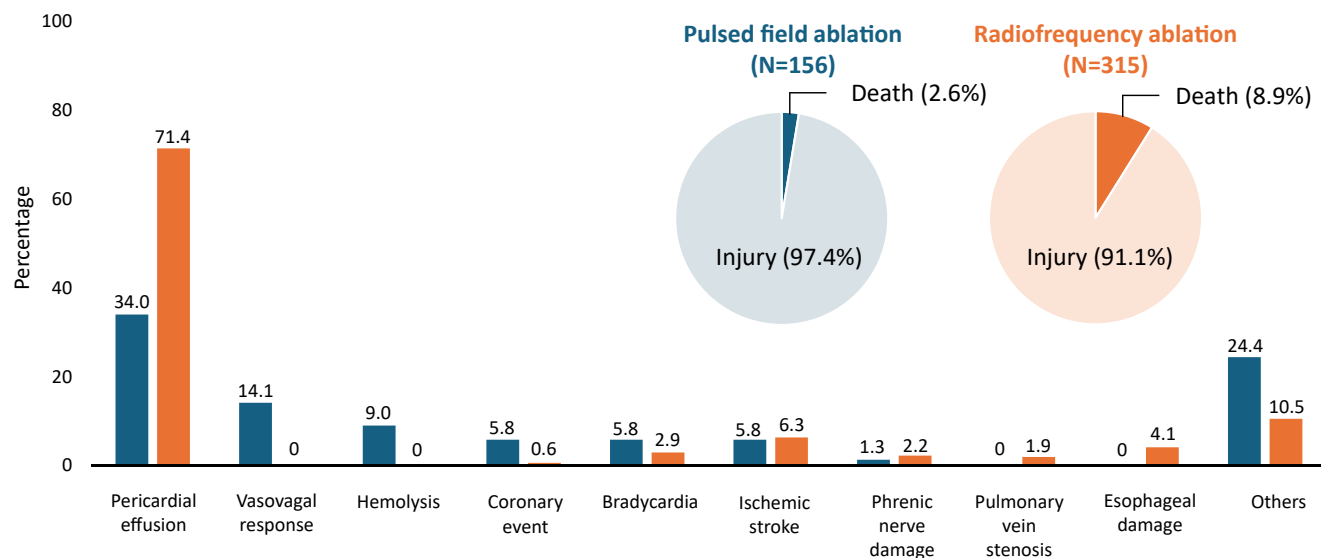
Proportion of reported events with PFA vs RF ablation

There were significant differences in the proportion of reported events for clinically important categories. The proportion of deaths reported as clinical complications was significantly higher with RF than with PFA (8.9% vs 2.6%; $P=.010$). The proportion of pericardial effusion was higher with RF use than with PFA (71.4% vs 34.0%; $P<.001$), though the proportion of patients requiring pericardiocentesis or surgical management did not differ significantly (95.6% vs 88.7% for RF and PFA, respectively; $P=.092$).

Vasovagal response (14.1% vs 0%), hemolysis (9.0% vs 0%), and coronary events (5.8% vs 0.6%) comprised a higher proportion with PFA than with RF ($P<.001$ for all). The overall proportion of bradyarrhythmias was not different (5.8% vs 2.9%

**Figure 1**

Study flow diagram. MAUDE = Manufacturer and User Facility Device Experience; PFA = pulsed field ablation; RF = radiofrequency.

**Figure 2**

Reported proportion of complications classified as "death" or "injury."

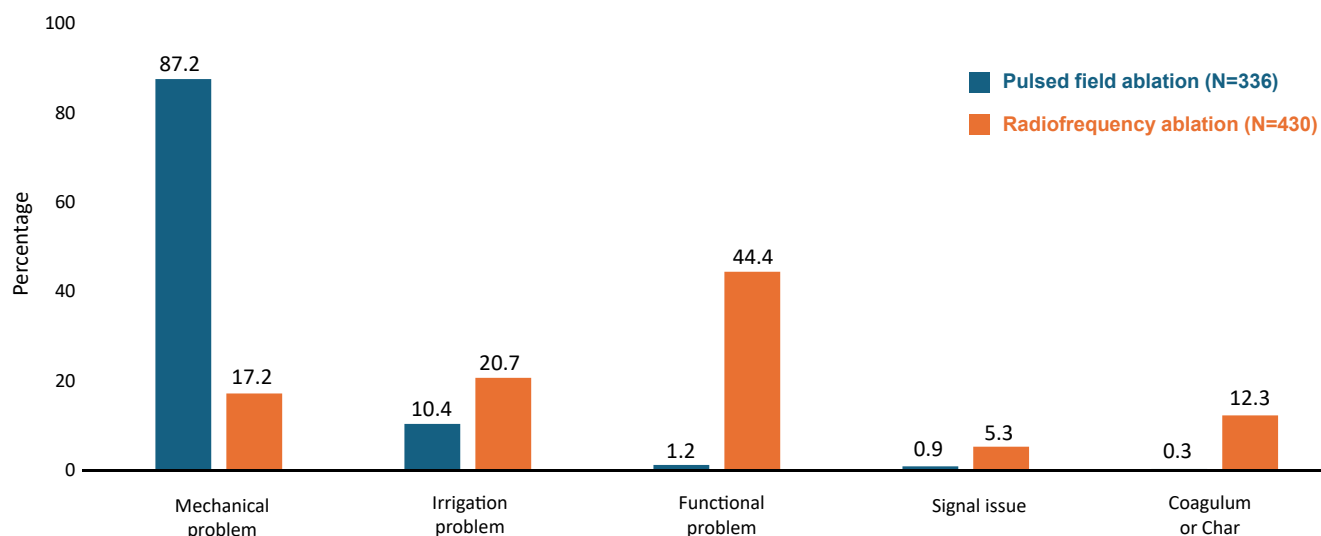
for PFA and RF, respectively; $P=.131$), though SND comprised a higher proportion of the reported events with PFA (3.8% vs 0.3%; $P=.006$). AVB requiring PPM was reported only with RF (1.5% vs 0%; $P=.176$), but this difference did not reach statistical significance. The proportions of ischemic stroke (5.8% vs 6.3% for PFA and RF, respectively; $P=.805$) and phrenic nerve injury (1.3% vs 2.2% for PFA and RF, respectively; $P=.724$) did not differ significantly between the groups. Pulmonary vein stenosis (1.9% vs 0%; $P=.184$) and esophageal damage (4.1% vs 0%; $P=.006$) were exclusively reported after RF use. Other reported clinical complications did not show significant differences between the groups.

Regarding reports of catheter malfunction, the proportion of mechanical problems was significantly higher after PFA use than after RF use (87.2% vs 17.2%; $P<.001$). The proportion of

irrigation or flushing issues was noted in both groups but was higher after RF use (20.7% vs 10.4%; $P<.001$). Functional problems (1.2% vs 44.4%), signal issues (0.9% vs 5.3%), and coagulum or char formation (0.3% vs 12.3%) were more frequently reported after RF use ($P<.001$ for all).

Discussion

The main findings of this study are as follows: (1) The most common clinical complications reported after PFA use were pericardial effusion, vasovagal response, and hemolysis. (2) The most common catheter malfunction after PFA use was mechanical dysfunction, specifically guidewire entrapment or deformity to the spline or array. (3) The reported rates of complications significantly differed between PFA and RF ablation.

**Figure 3**

Reported proportion of complications classified as "malfunction."

PFA has recently emerged as a potential paradigm-shifting technology for AF ablation. While pivotal randomized trials and prospective registries have demonstrated comparable efficacy and safety to conventional ablation energies, PFA carries significant advantages in procedural efficiency and avoidance of damage to adjacent thoracic structures.^{6–10} Notably, with the recent FDA approval of 2 PFA devices, Farawave and PulseSelect catheters for the treatment of AF, the use of PFA is expected to increase rapidly in US clinical practice. The goal of this analysis was to shed light on the distinct catheter-related complications associated with PFA, which differ considerably from those observed with RF catheters.

Recently, in the MANIFEST-17K study from MANIFEST-PF registry (Multi-National Survey on the Methods, Efficacy, and Safety on the Post-Approval Clinical Use of Pulsed Field Ablation), the safety of PFA was reported in more than 17,000 patients, detailing the incidence of individual complication and providing valuable clinical insights.¹² However, MANIFEST-17K was a carefully selected cohort, conducted predominantly at early adopting centers and included only those patients who met the inclusion/exclusion criteria, and therefore its findings could differ from general practice. In contrast, MAUDE includes mandatory report from the manufacturers, and the data may reflect a real-world practice pattern in an unselected patient population more closely. It is important to acknowledge that MAUDE is limited by underreporting of significant clinical complications.¹³ While manufacturers were required to report complications, reporting practices may vary across manufacturers. Thus, the limitations of the MAUDE database highlight an unmet need for an improved postmarket surveillance system of new devices. The Heart Rhythm Society EP PASSION collaborative (*ElectroPhysiology Predictable And SuStainable Implementation Of National Registries*) may offer an important alternative. However, despite the aforementioned limitations of MAUDE, the present analysis offers unique value in reporting adverse events in general practice in the unselected population that may be common or rare or those that clinicians may not be aware of and thus complements observations from prospective registries and trials.

The most intriguing PFA-specific complication was hemolysis. Since the initial report by Venier et al,¹⁴ multiple reports of hemolysis have been documented. One potential explanation for severe hemolysis is ablation of the blood pool as currently available PFA technologies lack high-fidelity mechanisms to confirm effective or uniform tissue contact, such as contact force sensors or impedance monitoring, which are standard in RF systems. The absence of reliable contact can lead to an increased number of energy applications and can result in hemolysis.¹¹ Although hemolysis was more prominently reported after PFA than after RF use,¹⁵ the degree of acute kidney injury and ablation time showed the strongest dose-response correlation with RF than with PFA.¹⁶ Therefore, it is possible that the provider's learning curve, the immature status of the catheter, or the technique for efficient energy delivery may be contributing factors to the increased hemolysis,

which should improve over time. For contemporary practice, as more than half of the hemolysis cases led to acute renal failure, with approximately 50% requiring hemodialysis, clinicians should be aware of this potentially catastrophic complication, and further advancements in the precise delivery of efficient ablation energy are needed.

Coronary artery spasm is another adverse event observed more frequently after PFA.¹⁷ The specific mechanism of this complication is still not fully understood, but a local surge of sympathetic nerve activation could be a one possible mechanism.¹⁸ Although reversible in all reported cases, 1 patient experienced incessant and potentially life-threatening ventricular tachycardia. Attention should be paid to energy delivery near critical areas, such as the cavotricuspid isthmus or mitral isthmus, and the prophylactic use of nitroglycerin should be considered to mitigate this risk.^{19,20}

Vasovagal events are not uncommon in AF ablation procedures. Although vasovagal responses were generally benign and reversible, their frequency during ablation merits caution, including preparations for temporary pacing. The presence of vagal ganglion plexuses near the pulmonary veins is well documented, and the left superior pulmonary vein is reported to be most susceptible to the vagal response.^{21,22} There is evidence that ablation at the right superior pulmonary antrum may reduce the risk of significant vasovagal response.²² In the cases reported in MAUDE, more than half of the patients required atropine or temporary pacing to stabilize hemodynamics. Although routine use of anticholinergic therapy (eg, glycopyrrolate) before PFA is being used in many electrophysiology laboratories, further study is necessary. Although PFA is reported to have tissue selectivity with resulting permanent damage only to the adjacent myocardium, there may be transient reversible stimulatory effects, different from the durable impact on ganglionated plexi seen with RF energy application.²³ The higher incidence of SND in our reports may also be explained by similar mechanism (or by vasospasm of the sinoatrial nodal artery) and should raise operator awareness during procedures.

Consistent with expectations based on preclinical evaluation, structural complications such as esophageal fistula or pulmonary vein stenosis were not reported after PFA use.¹² The proportion of reports of pericardial effusion was lower after PFA, though the use of rescue therapies were similar. Although there is potential for an R-on-T phenomenon with nongated PFA delivery, there were no reports of ventricular fibrillation induced by the delivery of PFA on the T wave. In addition, the overall mortality rate was lower in PFA cases than in RF cases. These findings suggest that PFA could offer improved safety compared with RF, provided that transient side effects are minimized through careful attention and preparation during the procedure.

The majority of PFA catheter dysfunction was mechanical in nature, particularly related to deformity or configuration disruptions. Notably, almost half of the reports involving the PulseSelect catheter described the array becoming "knotted" during the procedure. Given that current PFA catheters are considerably thicker than RF catheters, kinking

or knotting poses a risk for vascular injury and complications with hemostasis, although expansion of a femoral vein injury was reported in only 1 case in MAUDE. Close attention to catheter shape during ablation is critical, and damaged catheters should be replaced immediately to avoid further mechanical disruption. As PFA systems continue to evolve and become integrated into 3-dimensional mapping systems, functional issues similar to those encountered with RF catheters may emerge.²⁴

There are several limitations to our analysis. As the exact number of procedures were not reported in the MAUDE database, our report described only the relative proportion of reported complications, not their incidence. The primary value of these data is descriptive in nature. The MAUDE database was not designed for clinical studies, and the contents of the report were not consistent. Although manufacturer reporting is mandatory, physician reporting is voluntary, which may lead to underreporting of unrecognized complications, a known limitation of the MAUDE database. Even though manufacturer reporting is mandatory, the reporting behaviors may vary from manufacturer to manufacturer and may thus also influence the rate of reported adverse events. Some important complications that require active screening (ie, pulmonary vein stenosis) could also be underrecognized or underreported. The higher proportion of deaths in the RF group should be interpreted with caution, as RF is possibly used in more complex cases in practice and patients undergoing RF and PFA are likely to have different characteristics. As the description of the event was in a disorganized form, there was a possibility of varying proportion of adverse events depending on how they were handled. Some important details of the complications, such as location of the catheter at the time of vasospasm and recovery of phrenic injury, could not be identified.

Conclusion

The reported complications after PFA and RF ablation differ significantly. More awareness of PFA-specific complications, which were uncommon after traditional RF ablation, is crucial to ensuring the safe use of PFA catheters in clinical practice.

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Appendix

Supplementary data

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.hrthm.2024.10.063>.

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